Event Structure and the Emergence of Time from Phase-Overlap Geometry

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Abstract

Paper XXII established that the E_8 root system contains two conjugate copies of the 600-cell. The present paper investigates the consequence of allowing a relative phase rotation between these two copies.

A relative rotation parameter θ is introduced as a geometric degree of freedom, not a time coordinate. As θ varies, pairs of vertices from the two 600-cell copies enter and leave geometric alignment. An event is defined as a vertex pair that becomes admissible for at least one value of θ, and each event is assigned a birth angle: the first value of θ at which the pair becomes admissible.

The set of events carries a natural strict partial order given by birth-angle comparison. This ordering is proved to be irreflexive, transitive, and non-total. It is not assumed as a background temporal structure; it is derived from the phase-overlap structure of the dual 600-cell geometry. The paper interprets this order as an emergent temporal structure: temporal succession arises from the ordering of admissible configurations rather than from an external clock. A no-canonical-global-clock theorem shows that no unique real-valued total ordering is determined solely by the event structure. A worked example on two pentagons demonstrates the construction explicitly.

The paper does not derive general relativity. It establishes a first step: a causal-type event order derived from discrete closure geometry without assuming spacetime.

1 Introduction

Paper XXII [Smart, 2026c] showed that the E_8 root system contains two conjugate copies of the 600-cell (the double cover, 240 roots $\to$ 120 vertex pairs). Papers IX–X [Smart, 2026a,b] established that continuous geometry and a gravitational analogue can emerge from the constraint manifold structure. The present paper takes the next step: deriving a causal event structure — and with it, emergent time — from the phase-overlap geometry of the dual 600-cell system.

Core claim

Time is not assumed as a parameter. An emergent temporal structure arises as the partial ordering of admissible overlap configurations between two phase-rotated copies of the 600-cell constraint geometry.

Scope

This paper constructs the event structure and proves its ordering properties. It does not derive the Einstein field equations, define a metric, or construct full general relativity. Those are subsequent targets (Papers XXIV–XXV). The goal here is to show that temporal ordering can arise from geometry alone.

2 The Dual 600-Cell System

2.1 Base geometry

Definition 1 (600-cell vertex set)**.** *Let $H = \{v_1, \ldots, v_{120}\} \subset S^3 \subset \mathbb{R}^4$ be the 120 vertices of the 600-cell on the unit 3-sphere. The adjacency relation is: $v_i \sim v_j$ iff $|v_i - v_j| = 1/\varphi$ where $\varphi = (1 + \sqrt{5})/2$.*

The graph Laplacian is $L = 12\,I - A$ where A is the adjacency matrix (degree 12).

2.2 Dual copies with phase rotation

Definition 2 (Dual phase system)**.** *The dual 600-cell system consists of:*

- *$H_1 = H$ (the reference copy),*
- *$H_2(\theta) = R_\theta\, H$ where $R_\theta \in SO(4)$ is a rotation parameterised by θ,*

with θ a geometric parameter, not a time coordinate.

Remark 3. *The rotation R_θ acts on $\mathbb{R}^4$ and moves the vertices of H_2 on S^3. The parameter θ may be taken along any one-parameter subgroup of $SO(4)$; for definiteness, we use a left-isoclinic rotation $R_\theta(x) = q(\theta) \cdot x$ where $q(\theta) = (\cos\theta/2, \sin\theta/2, 0, 0)$ is a unit quaternion. The qualitative construction below requires only such a one-parameter family of relative configurations. Symmetry-related choices yield equivalent formulations; more general choices may alter quantitative details without changing the overall framework.*

3 Overlap and Admissibility

3.1 Overlap map

Definition 4 (Overlap)**.** *For vertices $v_i \in H_1$ and $w_j \in H_2(\theta)$, define the overlap function*

$$\Omega(v_i, w_j, \theta) = \begin{cases} 1 & \text{if } \|v_i - w_j(\theta)\| < \delta, \\ 0 & \text{otherwise,} \end{cases} \tag{1}$$

where $\delta > 0$ is a fixed alignment threshold satisfying $\delta < 1/\varphi$.

Remark 5. *The condition $\delta < 1/\varphi$ ensures that overlap is a strict proximity condition and does not coincide with adjacency in the 600-cell graph. Thus overlap detects alignment events rather than structural edges.*

3.2 Admissibility

Definition 6 (Admissible pair)**.** *A pair (v_i, w_j) at angle θ is* admissible *if*

$$\mathcal{A}(v_i, w_j, \theta) = \Omega(v_i, w_j, \theta). \tag{2}$$

Remark 7. *For the purposes of this paper, admissibility is defined purely by geometric alignment. More refined closure constraints may be introduced in later work; here we isolate the event structure generated by overlap alone.*

4 Events

Definition 8 (Event)**.** *An* event *is a pair (v_i, w_j) such that there exists at least one θ with*

$$\mathcal{A}(v_i, w_j, \theta) = 1.$$

Definition 9 (Birth angle)**.** *For an event $e = (v_i, w_j)$, define its* birth angle

$$\theta_b(e) = \inf\{\theta \in [0, 2\pi) : \mathcal{A}(v_i, w_j, \theta) = 1\}. \tag{3}$$

Remark 10. *The birth angle is the first geometric configuration at which a given vertex pair becomes admissible. Events are static geometric objects, each equipped with an intrinsic emergence parameter θ_b.*

Let

$$\mathcal{E} = \{(v_i, w_j) : \exists\theta \text{ with } \mathcal{A}(v_i, w_j, \theta) = 1\}$$

be the event set.

5 The Ordering Relation

5.1 Construction

Definition 11 (Event ordering)**.** *For events $e_1, e_2 \in \mathcal{E}$, define*

$$e_1 \prec e_2 \quad \textit{iff} \quad \theta_b(e_1) < \theta_b(e_2). \tag{4}$$

Remark 12. *The parameter θ is not interpreted as time. The ordering is defined entirely on the event set via the intrinsic emergence parameter θ_b.*

5.2 Properties

Proposition 13 (Strict partial order)**.** *The relation $\prec$ on $\mathcal{E}$ is a strict partial order: it is irreflexive, transitive, and non-total.*

Proof. **Irreflexivity:** For any e, $\theta_b(e) < \theta_b(e)$ is false, so $\neg(e \prec e)$.

Transitivity: If $e_1 \prec e_2$ and $e_2 \prec e_3$, then

$$\theta_b(e_1) < \theta_b(e_2) < \theta_b(e_3),$$

hence $\theta_b(e_1) < \theta_b(e_3)$ and therefore $e_1 \prec e_3$.

Non-totality: If $\theta_b(e_1) = \theta_b(e_2)$, then neither $e_1 \prec e_2$ nor $e_2 \prec e_1$ holds. Such events are incomparable. The pentagon example (Section 8) exhibits multiple events sharing the same birth angle. □

5.3 Interpretation

The partial order $\prec$ on $\mathcal{E}$ defines an emergent temporal structure.

Temporal succession arises from the ordering of admissible configurations. The ordering is not imposed externally; it is determined by the geometry of phase-overlap.

Remark 14. *The parameter θ is not interpreted as physical time. It is a coordinate on the space of relative configurations of the dual 600-cell system. The temporal structure arises not from θ itself, but from the order type induced by the emergence of admissible configurations. In particular, only the ordering of birth angles is used; no metric or duration is derived from θ.*

Remark 15. *An event represents the minimal admissible interaction channel between the two phase-rotated structures. In this formulation, events are defined at the level of vertex pairs; higher-order interaction structures (edges, faces, or incidence relations) may be incorporated in refined models.*

6 No Canonical Global Clock

Theorem 16 (No canonical global clock)**.** *There is no unique function $t : \mathcal{E} \to \mathbb{R}$ determined solely by $(\mathcal{E}, \prec)$ such that*

$$e_1 \prec e_2 \Rightarrow t(e_1) < t(e_2)$$

and which totally orders all events.

Proof. Since $(\mathcal{E}, \prec)$ is a non-total partial order, there exist incomparable events e_1, e_2 with neither $e_1 \prec e_2$ nor $e_2 \prec e_1$. Any total ordering extending $\prec$ must arbitrarily assign $t(e_1) < t(e_2)$ or $t(e_2) < t(e_1)$. Different choices yield distinct extensions. Hence no unique such function exists. □

Remark 17. *A total ordering consistent with $\prec$ (a linear extension) may be chosen, but it is not uniquely determined by the event structure. This reflects the absence of a preferred global temporal parameter.*

7 Observers

Definition 18 (Observer)**.** *An* observer *is a linear extension of* $(\mathcal{E}, \prec)$*, i.e., a total order* $\leq_O$ *such that*

$$e_1 \prec e_2 \;\Rightarrow\; e_1 \leq_O e_2.$$

Proposition 19 (Observer multiplicity)**.** *If a set of* k *events share the same birth angle, then there exist at least* $k!$ *distinct observers differing in their ordering of these events.*

Proof. All events with identical θ_b are mutually incomparable under $\prec$. Any permutation of these events yields a valid linear extension, giving $k!$ distinct observers. □

Remark 20. *This provides a combinatorial precursor to observer-dependent simultaneity: all observers agree on ordered events but may differ on the ordering of simultaneous (incomparable) events.*

8 Worked Example: Dual Pentagons

Example 21 (Two pentagons on S^1)**.** *Consider the 2D analogue: two regular pentagons on the unit circle* S^1*.*

Setup. *Pentagon* P_1 *has vertices at angles* $\alpha_k = 2\pi k/5$ *for* $k = 0, \ldots, 4$*. Pentagon* $P_2(\theta)$ *has vertices at* $\beta_l(\theta) = 2\pi l/5 + \theta$*.*

Overlap. *For threshold* $\delta = \pi/20$ *(much smaller than the vertex spacing* $2\pi/5$*), the overlap* $\Omega(\alpha_k, \beta_l, \theta) = 1$ *iff* $|\alpha_k - \beta_l(\theta)| < \delta$ *(mod* 2π*).*

Admissibility. *All overlapping pairs are admissible in this simplified model (no closure constraint beyond proximity).*

Events. *As* θ *increases from 0 to* $2\pi/5$*, the alignment pattern evolves:*

- *Events born at* $\theta = 0$*:* $\{(k,k) : k = 0, \ldots, 4\}$*, each with birth angle* $\theta_b(k,k) = 0$*. These are simultaneous: 5 mutually incomparable events.*
- $\theta \in (0, 2\pi/5 - \delta)$*: No new pairs become aligned. No new events born.*
- *Events born at* $\theta = 2\pi/5$*:* $\{(k, k{-}1) : k = 0, \ldots, 4\}$*, each with birth angle* $\theta_b(k, k{-}1) = 2\pi/5$*. Again 5 simultaneous events.*

Ordering. *The 5 events born at* $\theta_b = 0$ *all precede the 5 events born at* $\theta_b = 2\pi/5$ *(since* $0 < 2\pi/5$*). Within each group of 5, no ordering exists (all share the same birth angle). The partial order has two "layers":*

$$\{(0,0),(1,1),(2,2),(3,3),(4,4)\} \;\prec\; \{(0,4),(1,0),(2,1),(3,2),(4,3)\}$$

No global clock. *At* $\theta = 0$*, the 5 events are incomparable, so any total ordering of them is an observer choice. There are* $5! = 120$ *distinct observers at this layer alone.*

Analogy to the 600-cell. *The pentagon has 5 vertices and* φ*-geometry (diagonal/side* $= \varphi$*). The 600-cell is the 4D generalisation with 120 vertices. The event structure scales accordingly: at each discrete alignment angle, up to 120 simultaneous events occur, and the partial ordering between alignment angles gives the emergent temporal structure.*

Table 1: Comparison with causal set theory.

Feature	VFD event structure	Causal set theory
Events	Admissible vertex pairs	Fundamental element
Ordering	Derived from geometric admissibility (via birth angles)	Assumed as causal re
Origin of set	Derived from 600-cell geometry	Postulated
Discreteness	From vertex structure	Postulated
Partial order	Proved (Prop. 13)	Assumed
Preferred global clock	Not canonically determined (Thm. 16)	Not fundamental
Observer	Linear extension of $\prec$	Choice of compatible

9 Relation to Causal Set Theory

The event structure $(\mathcal{E}, \prec)$ has formal similarities to causal set theory:

Key distinction: In the VFD framework, the event set and its ordering are *derived* from the constraint geometry. In causal set theory, they are postulated as fundamental. The VFD approach provides a geometric origin for the causal structure that causal set theory takes as given.

10 Discussion

10.1 What is established

- The dual 600-cell system with phase rotation is explicitly defined (Definitions 1–2).
- Overlap and admissibility are explicitly constructed (Definitions 4–6).
- Events are defined as admissible overlap configurations (Definition 8).
- The ordering relation $\prec$ is proved to be a strict partial order (Proposition 13).
- The no-global-clock theorem is proved (Theorem 16).
- Observers are defined as linear extensions, with multiplicity proved (Proposition 19).
- A worked example demonstrates all constructions explicitly.

10.2 What is not established

- No metric is derived. The partial order gives temporal ordering but not durations.
- No Einstein equations. The curvature-gravity connection of Papers IX–X is not upgraded to field equations.
- No Lorentz invariance. The event structure has discrete symmetry (H_4), not continuous Lorentz symmetry.
- No spatial geometry. The present paper addresses only temporal ordering; spatial structure requires further work.

10.3 Connection to the programme

The dual 600-cell system of this paper is the same conjugate pair identified in Paper XXII. The phase rotation θ is the geometric degree of freedom between Copy A and Copy B. The events are the discrete configurations at which the two copies are in admissible alignment. The temporal ordering is the structural consequence of the geometry varying with θ.

This completes the connection:

- Papers I–V: spectral structure (what the constraint geometry produces),
- Papers XII–XXI: quantum dynamics (how it evolves),
- Paper XXII: the spectral bridge (how everything connects),
- Paper XXIII: event structure and emergent time (the first step toward gravity).

11 Conclusion

Time is not assumed in the VFD closure framework. An emergent temporal structure is realised as the birth-angle ordering of admissible overlap configurations between two phase-rotated copies of the 600-cell constraint geometry. The ordering is irreflexive, transitive, and non-total (proved). No canonical global clock is determined by the event structure (proved). Different observers correspond to different linear extensions of the partial order, agreeing on ordered events but potentially disagreeing on simultaneous ones.

This is the first step toward gravity within the VFD programme. The next steps — metric structure (Paper XXIV) and field equations (Paper XXV) — will build on this event structure. The present paper establishes only the temporal ordering, not the full spacetime geometry.

No external time variable, spacetime manifold, or metric is assumed. The temporal structure arises entirely from the ordering of admissible configurations within the dual 600-cell geometry. This establishes a minimal geometric origin for causal ordering. Subsequent work will address the emergence of metric structure and dynamical laws on top of this event framework.

References

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